Ethanol Blending in Gasoline - Honduras

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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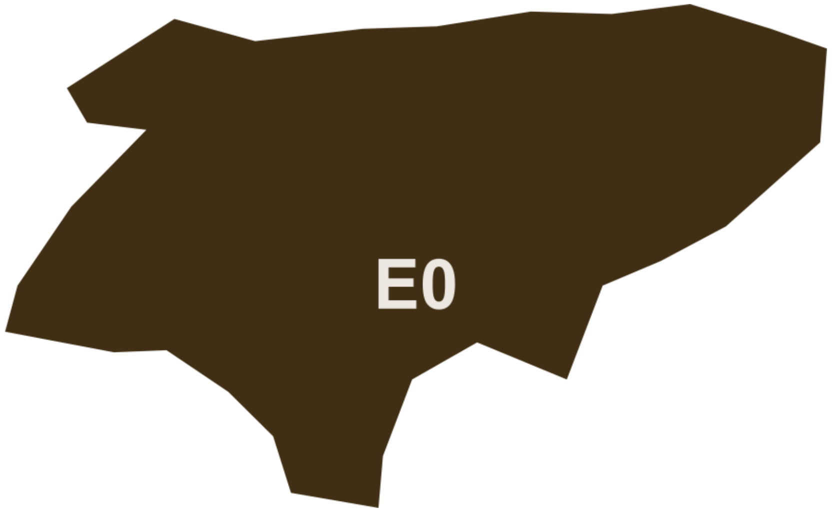
Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

Ethanol Blending in Gasoline - Honduras

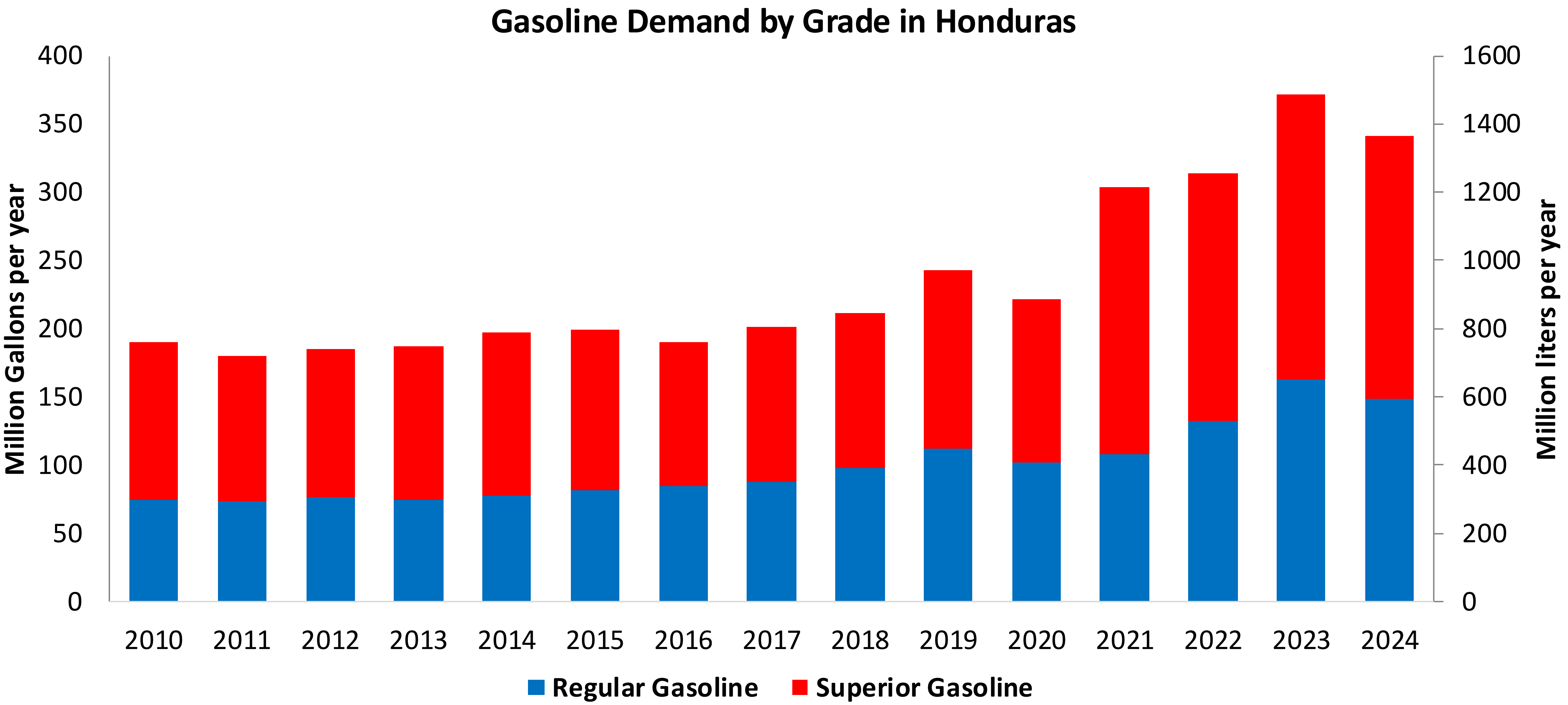


In 2024, gasoline demand was 374 million gallons (1,290 million liters). Regular gasoline RON 91 represented 46% while Superior gasoline RON 95 reached 54% of national demand. Honduras does not have gasoline production capacity and supplies its market with imports mainly from the United States.

Honduras has no national mandate for ethanol blends. A new biofuel's law is under discussion.

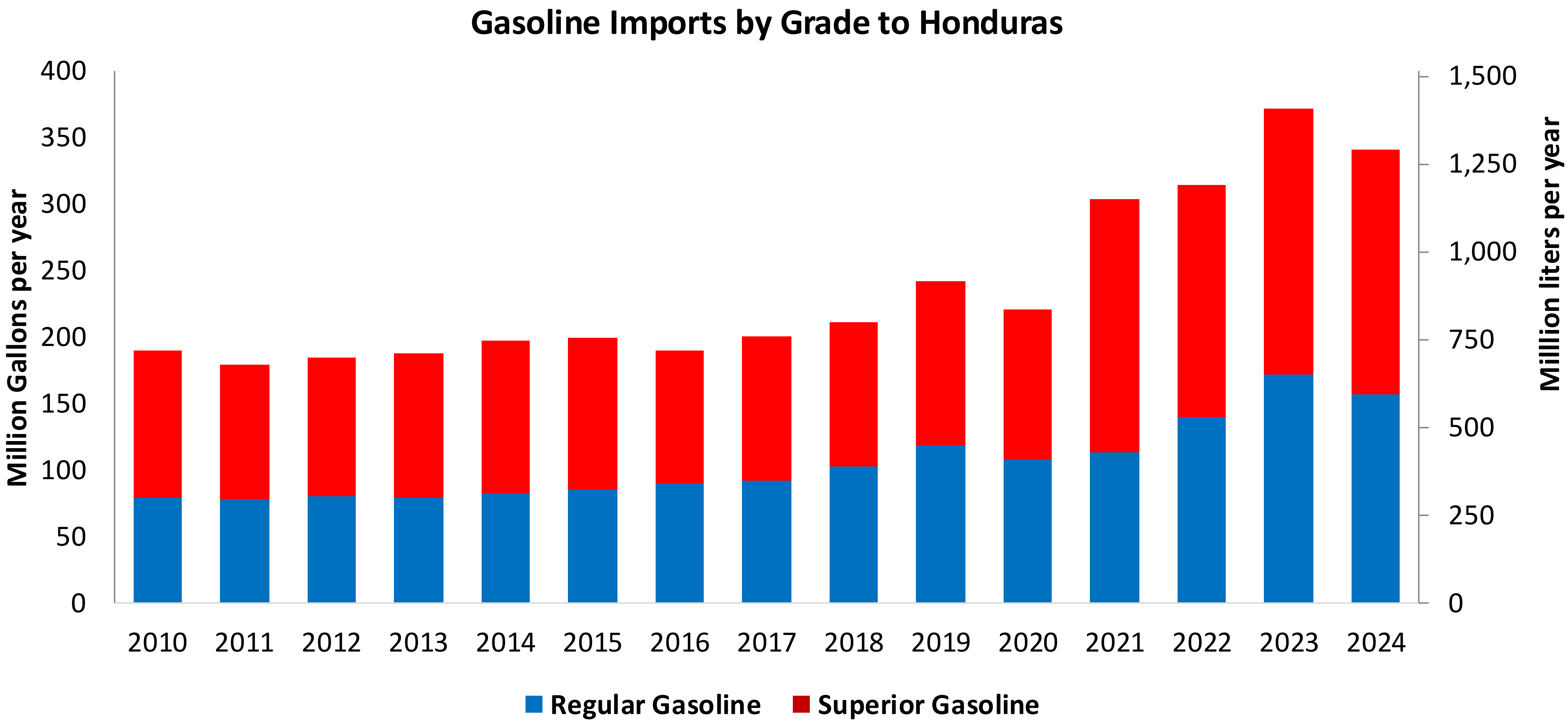
Source: CEPAL, Secretaría de Energía Honduras

Gasoline Demand in Honduras



Source: CEPAL, Secretaría de Energía Honduras

Gasoline Imports in Hondures



Source: CEPAL, Secretaría de Energía Honduras

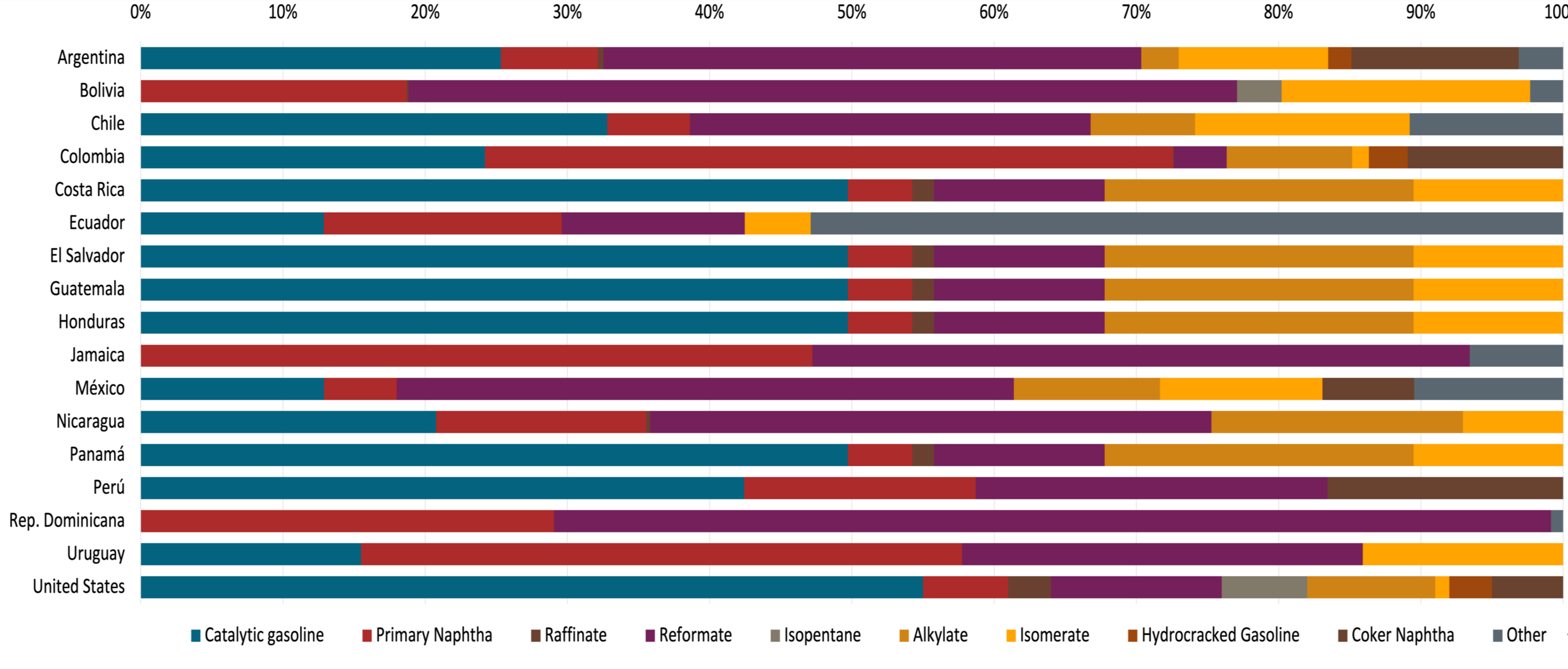
Gasoline Quality in Honduras

Name	RTCA 75.01.08:19 p.m.	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	2,5% v/v	2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0,25% v/v	0,25% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source:
RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



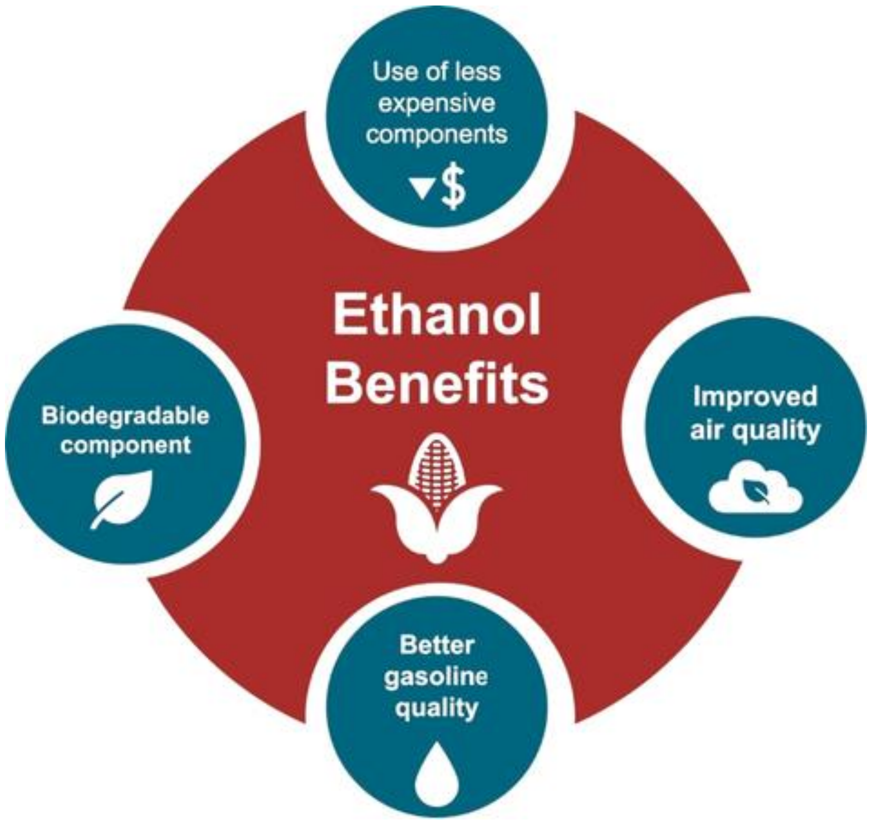
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

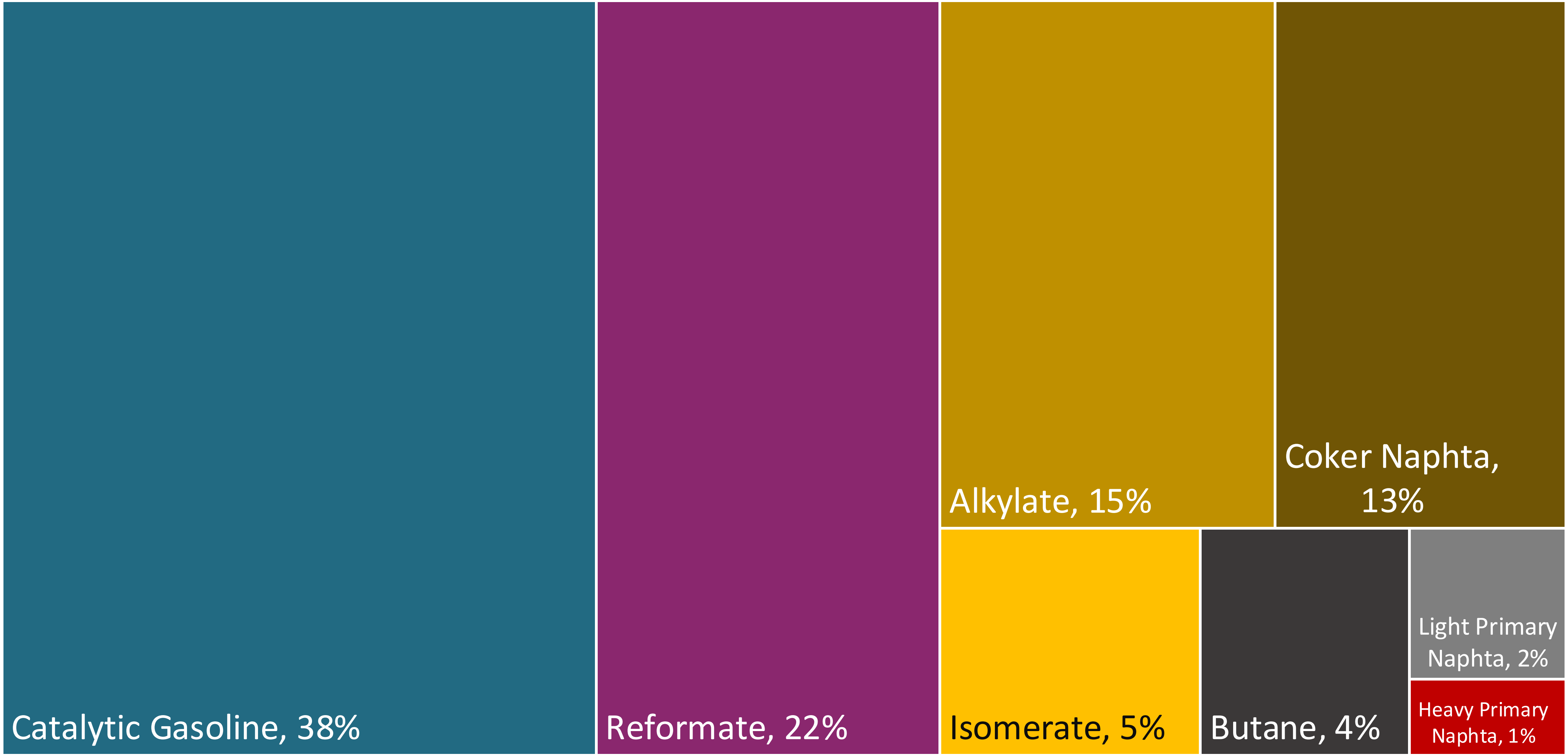
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

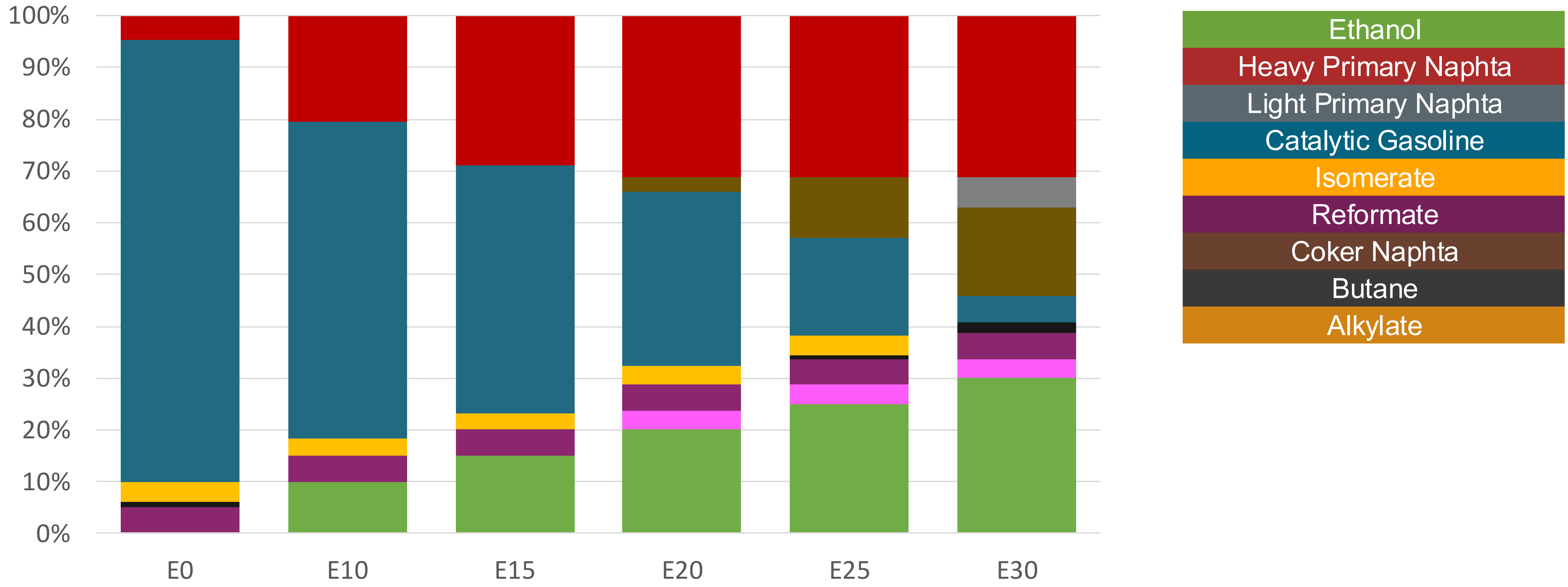


Blending Components - Honduras



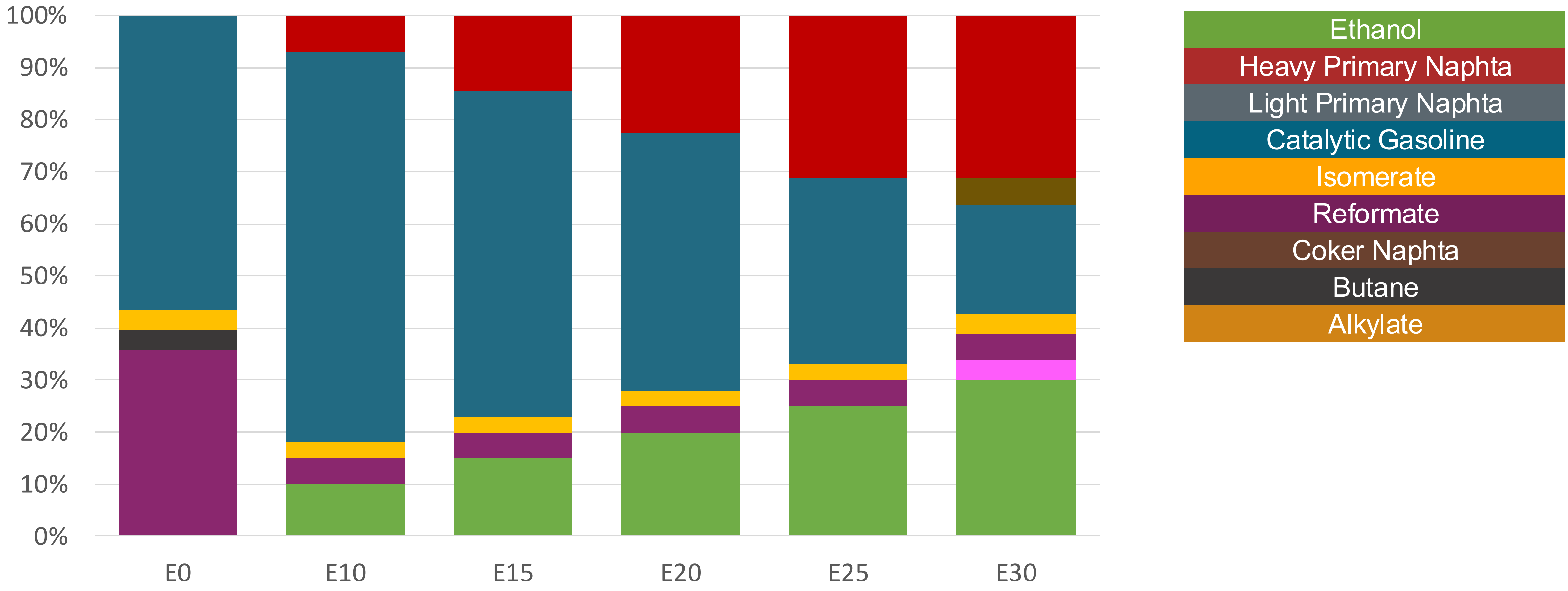
Source: Faro90

Honduras – Regular – Constant Octane



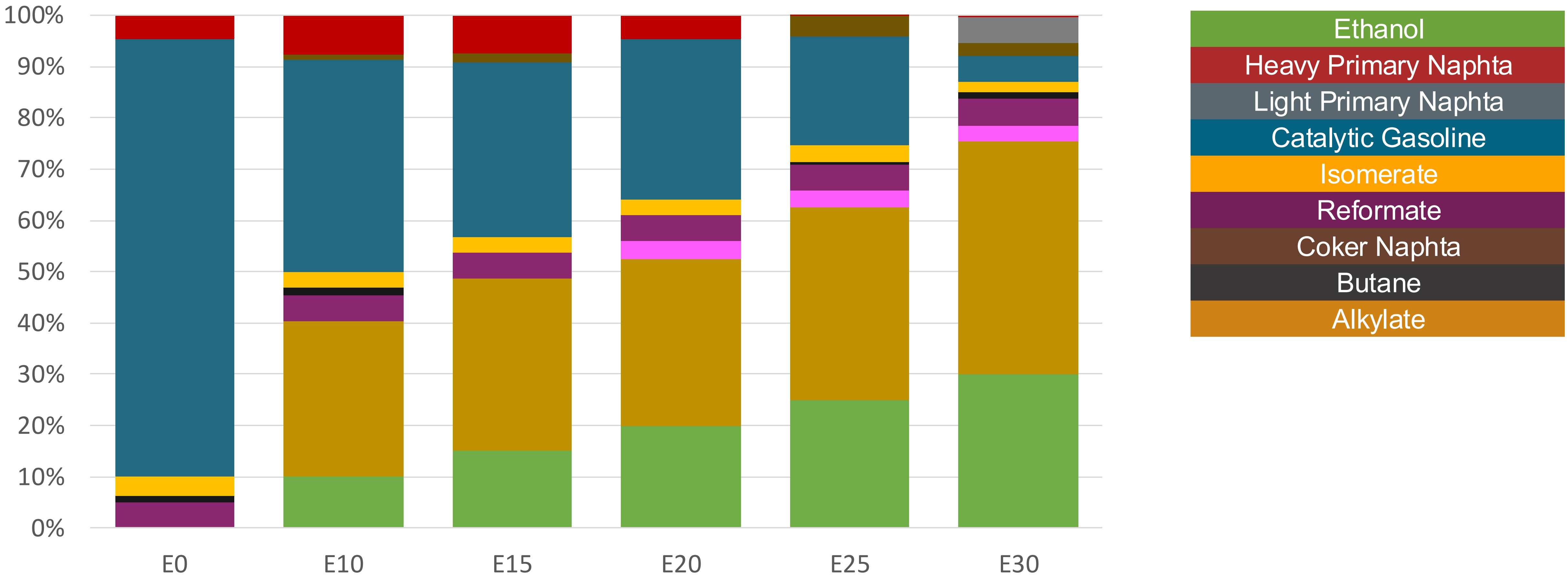
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.156	$ 1.885	$ 1.734	$ 1.592	$ 1.447	$ 1.477

Honduras – Premium – Constant Octane



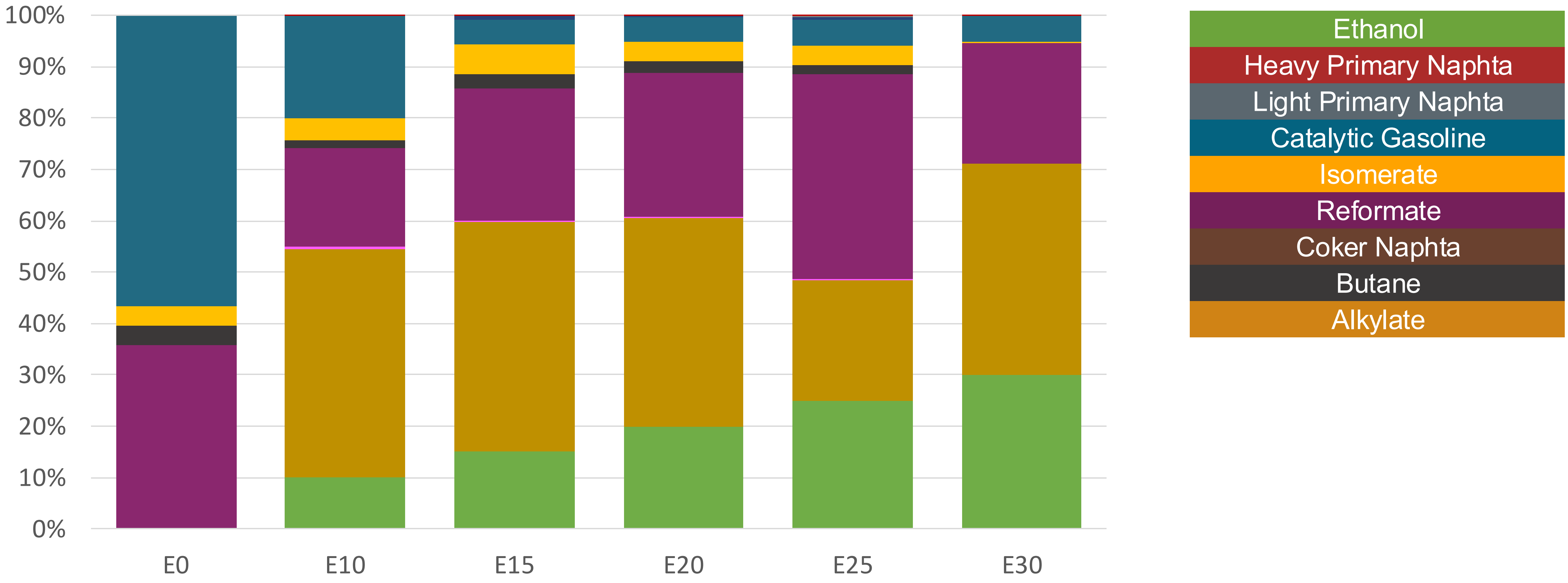
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Honduras – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156

Honduras – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

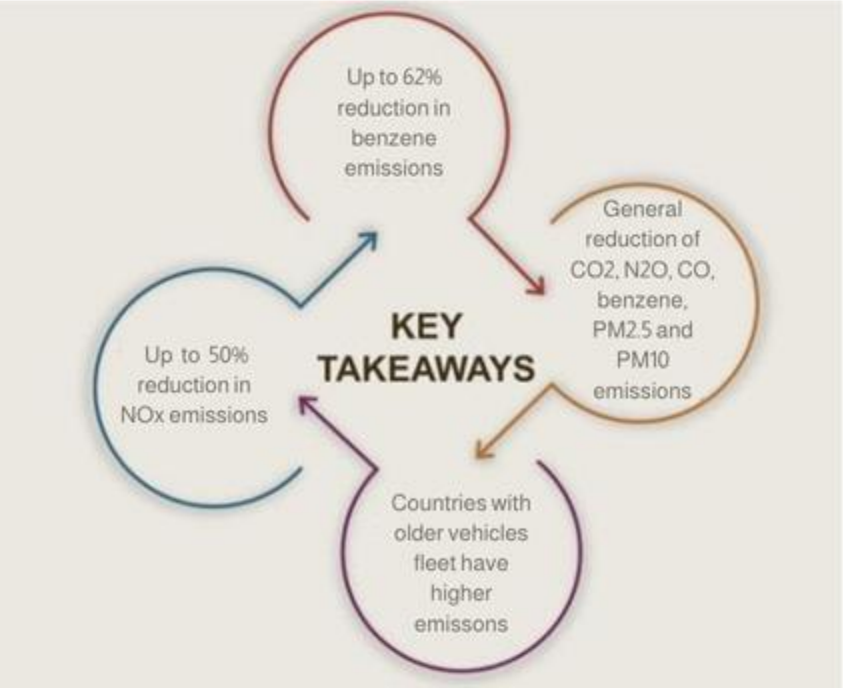
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicular Fleet in Honduras

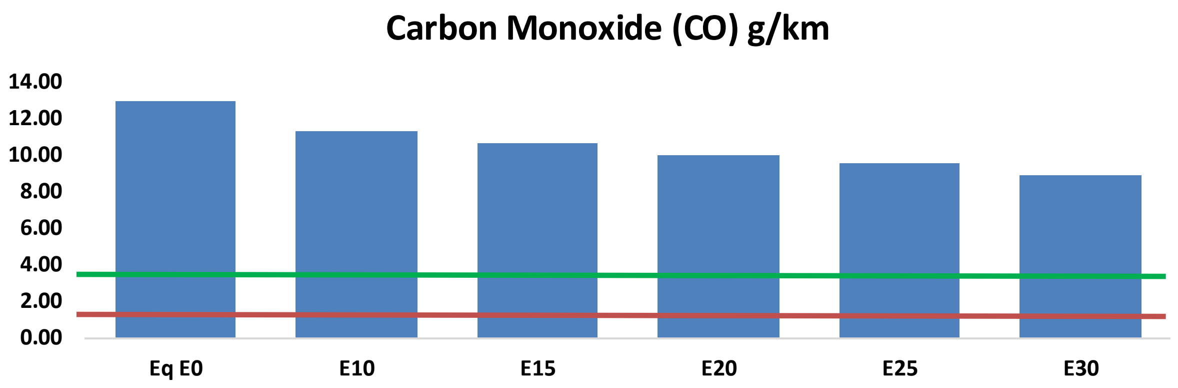
Type	Motor	Exhaust	Recuperator	Age	Vehicle Numbers
Light Vehicles	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	7,426
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	26,041
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	183,333
		Euro III		<8 years, < 160 mkm	121,208
		Euro IV		<8 years, < 160 mkm	81,137
		Euro V / Euro 6		<8 years, < 160 mkm	49,480
				4-8 years, 80-160 mkm	26,632
				>4years, >80 mkm	13,759
Medium Size Vehicles	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	18,844
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	165,407
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	364,696
		Euro III		<8 years, < 160 mkm	155,857
		Euro IV		<8 years, < 160 mkm	90,278
		Euro V / Euro 6		<8 years, < 160 mkm	67,324
				4-8 years, 80-160 mkm	65,322
				>4years, >80 mkm	92,978
Small Engines	FI 4-ciclos	None	PCV	<8 years, < 160 mkm	29,558
		Euro II		<8 years, < 160 mkm	35,954
		Euro III		<8 years, < 160 mkm	386,853
		Euro IV		<8 years, < 160 mkm	84,724
				4-8 years, 80-160 mkm	289,951
		Euro V		4-8 years, 80-160 mkm	120,241
				>4years, >80 mkm	864,831

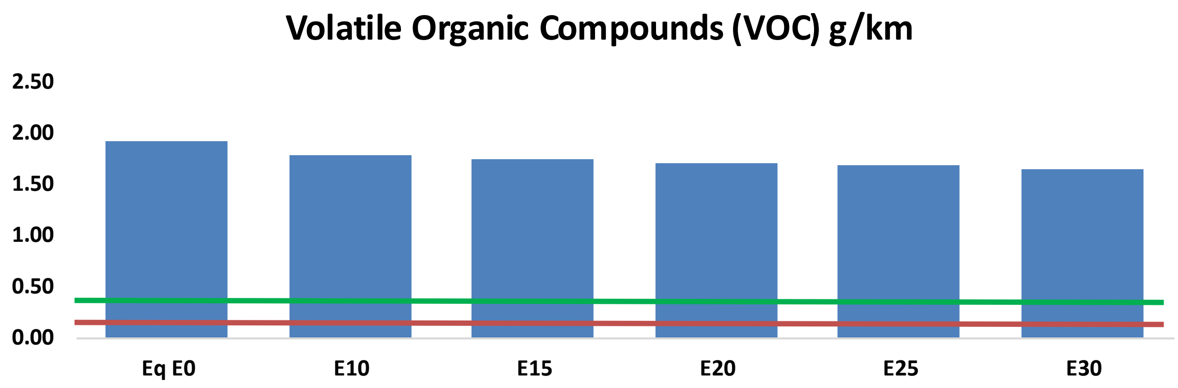
Vehicular Fleet: **3,341,832**
Average Age: **13 years**
Motorcycles: **50%**

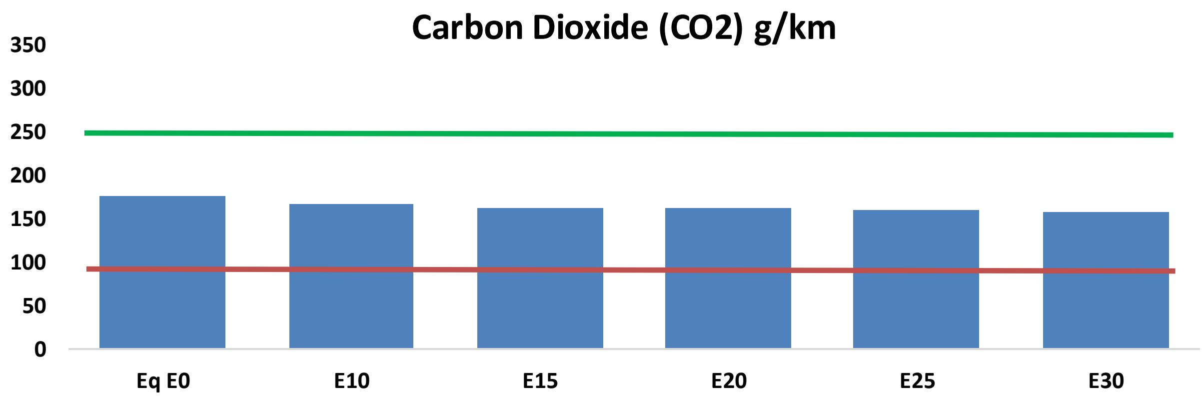
Source: INE, analysis Faro 90

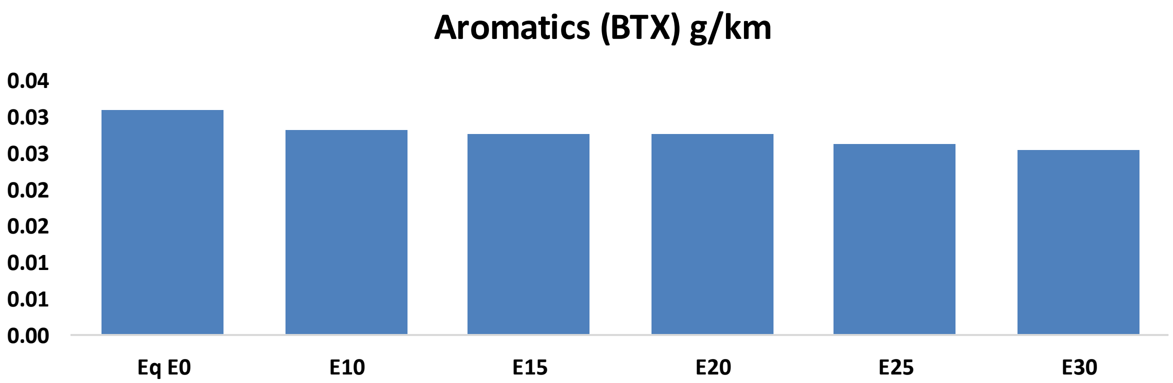
Honduras – Vehicular Emissions

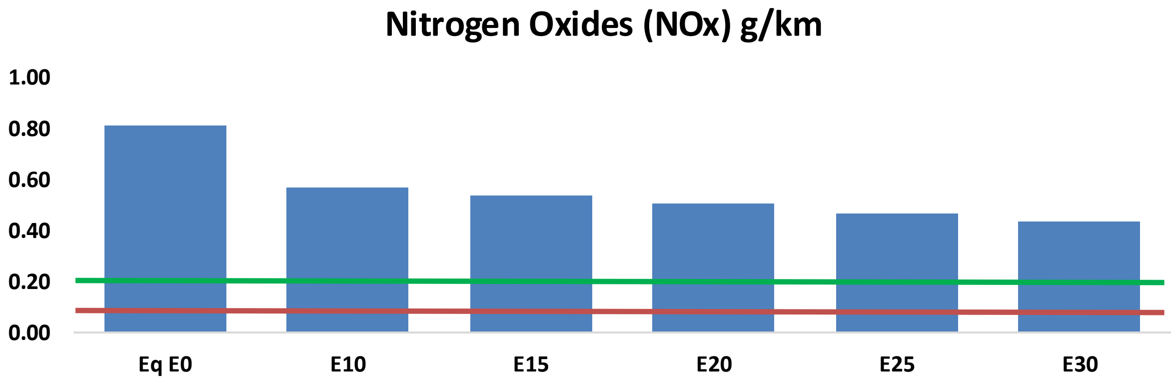
United States
Euro 6

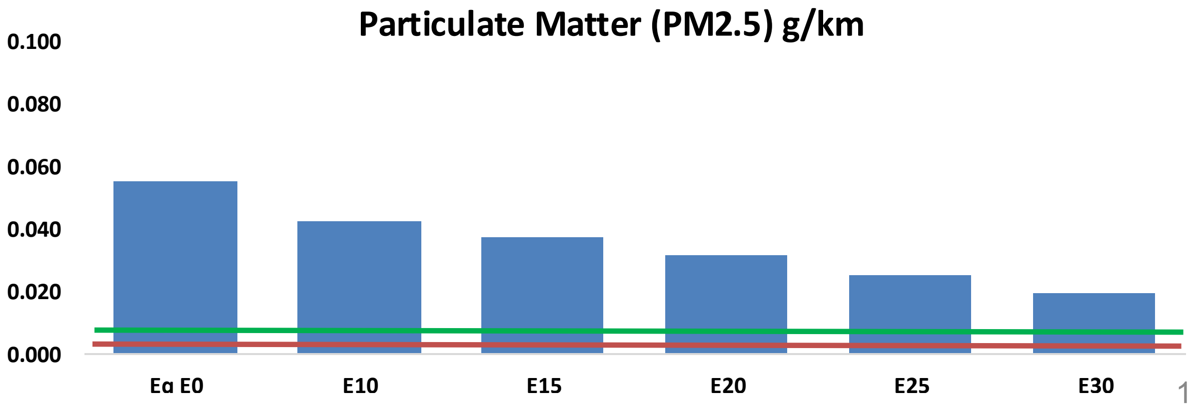












Source: Faro90

Honduras – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	769,836	720,927	672,017	633,942	596,549	568,074	532,699	-13%	-23%
CO2	10,458,718	10,189,779	9,935,782	9,736,021	9,637,171	9,612,562	9,435,375	-5%	-8%
VOC	114,312	110,202	106,091	103,636	101,585	100,134	97,751	-7%	-11%
NOx	48,124	36,093	33,687	31,750	29,943	27,951	25,843	-30%	-38%
SOx	128	119	109	101	93	84	75	-15%	-28%
NH3	4,192	4,150	4,109	4,128	4,175	4,208	4,235	-2%	0%
N2O	193	192	191	192	196	198	200	-1%	2%
CH4	25,506	25,506	25,506	25,889	26,450	26,934	27,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	434	406	378	356	334	318	296	-13%	-23%
Acetaldehyde	1,088	1,249	1,833	2,791	3,802	4,345	5,134	68%	249%
Formaldehyde	4,118	4,004	4,667	5,480	5,741	6,351	6,914	13%	39%
Benzene	1,849	1,776	1,683	1,657	1,643	1,571	1,520	-9%	-11%
PM2.5	653	580	507	440	374	303	230	-22%	-43%
PM10	2,626	2,332	2,037	1,769	1,502	1,219	924	-22%	-43%

Source: Faro90

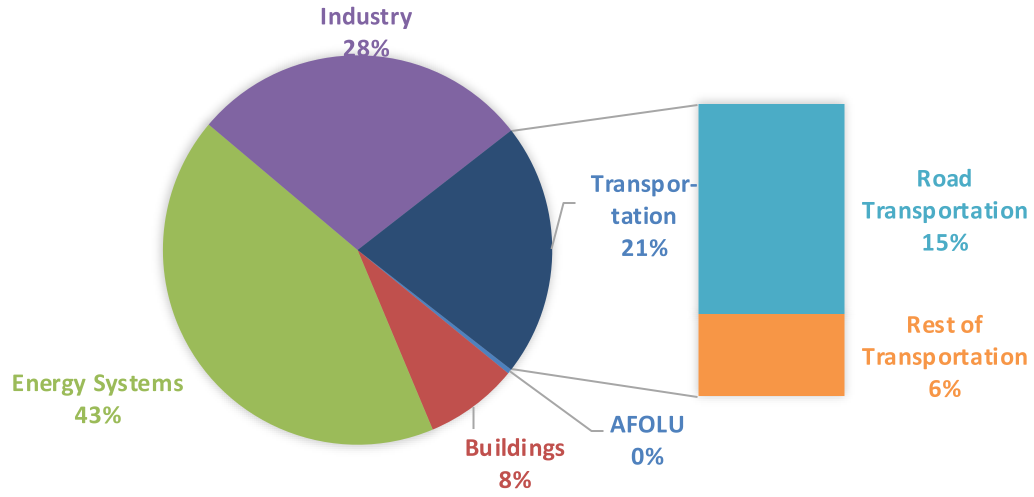
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

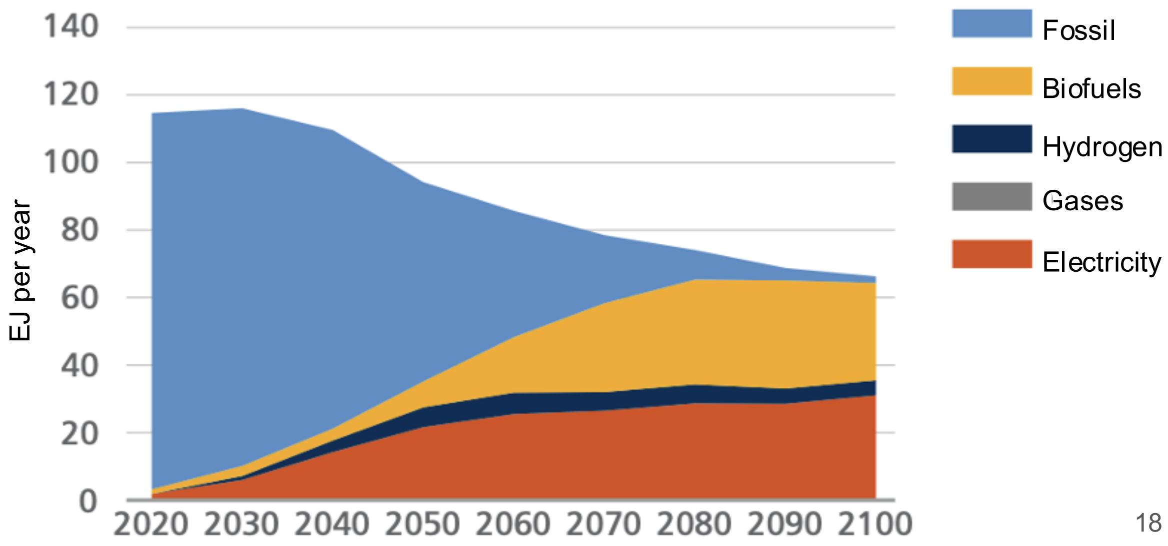
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

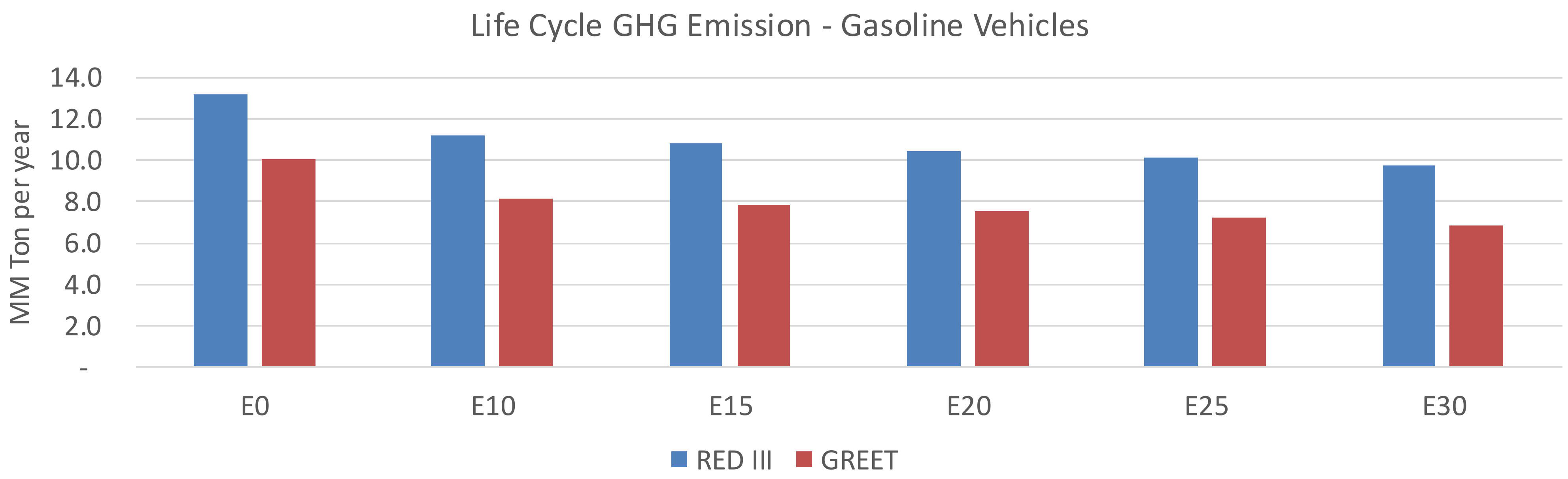
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Honduras – Gasoline Vehicle Fleet GEI Emissions



Emisiones País 2024 (MMT/año)	30.7
Meta a 2035 (MMT/año)	15.3
Delta	15.3

%Reducción vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	18.7%	22.2%	25.1%	28.0%	31.6%
RED II/III	0.0%	15.1%	18.0%	20.6%	23.1%	26.2%

% Meta@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.3%	14.5%	16.5%	18.4%	20.7%
RED II/III	0.0%	12.9%	15.5%	17.7%	19.9%	22.5%

Source: Greet, RED II/III, climateactiontracker.org, F90